ISESE
FORM TWO
PHYSICS MARKING SCHEME

SECTION A

1. 10 marks

I	II	III	IV	V	VI	VII	VIII	IX	X
B	C	B	C	A	B	C	B	B	C

2. 5 marks

I	II	III	IV	V
D	B	E	A	C

SECTION B (70 MARKS @10 MARKS)

3(a) (i) Distinction:

- **Scientific Theory:** A comprehensive, well-substantiated explanation of an aspect of the natural world that is gathered through the scientific method and repeatedly tested and confirmed through observation and experimentation.
- **Scientific Principle/Law:** A precise statement or description of an observed natural phenomenon that always happens under specific

conditions, often formulated mathematically, without necessarily explaining the underlying mechanism.

- **(a) (ii) Progression:**
  - An initial hypothesis undergoes rigorous, repeatable testing and controlled experimentation. When different independent scientists consistently obtain identical results that support the hypothesis without contradiction, it graduates to a theory and eventually becomes accepted as a universal scientific principle or law.
- **(c) (i) First Step:**
  - Observation.
- **(ii) Hypothesis:**
  - Presence of moisture/dampness accelerates the rate of rusting on metal tools. *(Accept any reasonable, testable hypothesis relating moisture to rust)*.
- **(iii) Variables:**
  - **Independent Variable:** The storage environment conditions (moisture content/humidity level).
  - **Dependent Variable:** The rate of rusting (formation of the reddish-brown layer).

4. (i) Vertical Forces:

- Weight (downward gravitational pull of the Earth).
- Normal reaction force (upward supporting force exerted by the floor).

- **(ii) Reason for Lack of Movement:**
  - The horizontal pushing force applied by the first student was smaller than or equal to the maximum static frictional force acting between the rough floor and the heavy crate, resulting in a net horizontal force of zero.
- **(iii) State of Forces:**
  - Static Equilibrium (or balanced forces).
- **(iv) Net Force Transition:**
  - When the second student joins, the combined horizontal force exceeds the maximum limiting static friction. This creates a non-zero **net unbalanced force** in the forward direction, causing the crate to accelerate from rest. Once uniform motion is established, the applied force balances the kinetic friction exactly, returning the net force to zero.
- **(v) Classification:**
  - **Contact Forces:** Pushing force, Frictional force, Normal reaction force.
  - **Non-contact Forces:** Weight (Gravitational force).

5. (a) (i) Relationship:

  - The weight of the floating block of wood is equal to the upward upthrust acting on it.
- **(a) (ii) Comparison:**
  - The weight of the displaced water is exactly equal to the total weight of the floating wood.
- **(a) (iii) Behavior in Denser Liquid:**
  - The block of wood would float higher in the liquid, meaning a smaller portion of its volume would be submerged. This is because a smaller volume of a denser liquid is required to displace a weight equal to the weight of the wood.
- **(b) (i) Governing Law:**
  - Hooke's Law.
- **(b) (ii) Beyond Elastic Limit:**
  - The spring undergoes permanent plastic deformation. It will no longer return to its original length when the stretching force is released, and its internal molecular structure becomes permanently distorted until it reaches the breaking point.

6. (a) (i) Wall Thickness Reason:

  - Liquid pressure increases directly with depth. Since the pressure is greatest at the very bottom of the dam, the lower walls must be built thicker and stronger to withstand the massive hydrostatic force exerted by the water.
- **(a) (ii) Density Effect:**
  - Internal liquid pressure is directly proportional to the density of the liquid. A higher liquid density results in greater pressure at any given depth.
- **(a) (iii) Deep-Sea Diving Suits:**
  - At extreme ocean depths, the surrounding water pressure is immense. The reinforced diving suits protect the human body from being crushed by resisting this extreme external hydrostatic pressure.
- **(b) (i) State of Acceleration:**
  - The acceleration of the car is zero because it is moving at a constant speed along a straight, flat highway, meaning there is no change in its velocity.
- **(b) (ii) Distance vs. Displacement:**
  - **Distance:** The actual total path length covered by the moving car, which is a scalar quantity (has magnitude only).

- **Displacement:** The linear distance measured from the starting point to the final position in a specific direction, which is a vector quantity (has both magnitude and direction).

7. (a) (i) Justification:

- In physics, work is defined as the product of force and displacement in the direction of the force ($W = F \times d$). Since the barbell remains completely stationary, its displacement is zero ($d = 0$), which means no mechanical work is done on the barbell.

- **(a) (ii) Energy vs. Power:**
  - **Energy:** The capacity or ability of a system to perform work.
  - **Power:** The rate or speed at which that work is done or energy is transformed over time.
- **(a) (iii) Form of Energy:**
  - Gravitational Potential Energy.
- **(b) (i) Energy Transformation:**
  - Solar energy (Light energy) → Electrical energy.
- **(b) (ii) Factor Reducing Efficiency:**
  - Accumulation of dust, dirt, or debris on the surface of the solar panel; cloud cover/shading; or high ambient temperatures.

8. (a) (i) Circuit Arrangement:

- Series Circuit.

- **(a) (ii) Main Disadvantage:**
  - If one appliance breaks, burns out, or is switched off, the entire circuit is broken. This cuts off current flow to all other electrical appliances in the household, causing them to go off simultaneously.
- **(a) (iii) Detection Instrument:**
  - Galvanometer *(Accept Ammeter)*.
- **(b) (i) Magnetization Method:**
  - Single Stroke Method.
- **(b) (ii) Testing Procedure:**
  - Bring the magnetized iron nail close to small magnetic materials like iron filings or a magnetic compass needle. If the nail attracts the filings or causes a noticeable deflection of the compass needle, it has successfully acquired magnetic properties.

9. (a) (i) Image Characteristics (Any two):

- Virtual
- Upright (Erect)

- Same size as the object
- Laterally inverted
- Image distance inside the mirror equals object distance in front of the mirror

- **(a) (ii) Reversal Reason:**
  - This is caused by the phenomenon of lateral inversion, where the plane mirror reverses the left and right sides of the object in its reflection.
- **(a) (iii) Reflection Phenomenon:**
  - Diffuse reflection (or irregular/unpolished reflection).
- **(b) (i) Apparent Color:**
  - Black.
- **(b) (ii) Reason:**
  - A red dress appears red because it reflects only red light and absorbs all other colors of the spectrum. When illuminated by pure blue light, there is no red component for the dress to reflect. The blue light is completely absorbed, leaving no reflected light to reach the observer's eyes.

SECTION C (15 MARKS)

10. (a) Charge Accumulation Cause:

- As the fuel truck travels at high speed, continuous friction between the metallic body of the truck and air molecules (air resistance), alongside internal friction from the sloshing petroleum liquid, causes a significant buildup of static electric charges on the outer chassis.

- **(b) Potential Danger:**
  - If these static charges accumulate without a path to escape, they can create a sudden, high-voltage electrostatic spark. Because the area surrounding a fuel tanker contains highly volatile and flammable petroleum vapors, this spark can easily ignite the vapors, causing a massive explosion or fire.
- **(c) Role of Dragging Chain:**
  - The heavy metal chain acts as an electrical conductor providing a continuous low-resistance path to the ground (earthing/grounding). It allows the accumulated static electricity to safely discharge into the asphalt road surface as fast as it forms, preventing dangerous charge buildup.
- **(d) Recommended Safety Practices (Any two):**
  - Always connect a dedicated grounding/earthing cable to the fuel tanker before starting any fuel offloading operations.

- Ensure vehicle engines are completely turned off near fuel discharge zones.
- Strictly prohibit the use of mobile phones or any other non-intrinsically safe electronic devices around fuel pumps.
- Use anti-static clothing/footwear to minimize personal charge generation when handling fuel lines.